

COMP 3700
Software Modeling & Design

Chapter2: Class Diagram
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Introduction

- The class diagram is fundamental to object-oriented programming
- UML's class diagrams capture the attributes and operations of each class as well the relationships that exist between classes

Class Icon

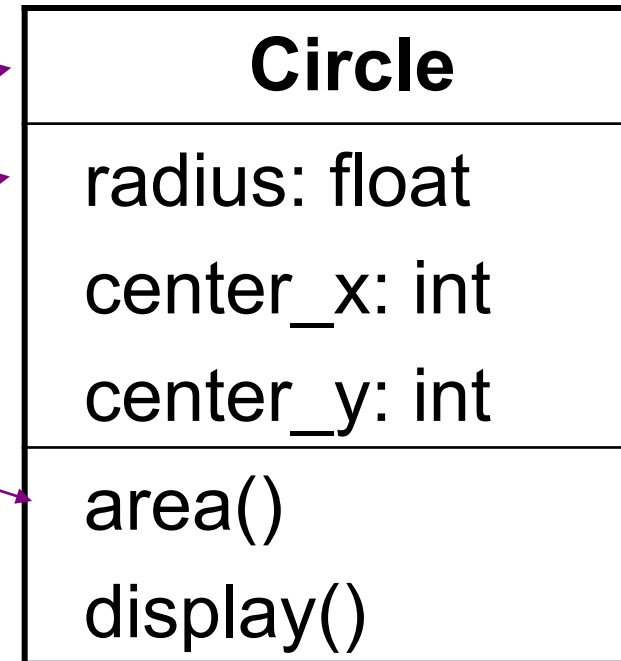
- Class Icon
- Hiding Details
- Visibility Notation
- Attribute Specification
- Operation Specification

Class Icon

- The UML class icon is a rectangle with three compartments:

- class name
- class attributes
- class operations

- Attributes are specified in the following form:
 - object:class name



Hiding Detail

- You can optionally leave out the attributes, operations, or both in a class icon:

Circle
area() display()

Circle
radius: float center_x: int center_y: int

Circle

UML Member Visibility Notation

- UML has three visibility prefixes for members:
+ for public, # for protected, and – for private

–e.g.

Circle
-radius: float
#area() +display()

Full UML Attribute Specification

- The full form of a UML attribute is as follows:

[visibility] name [multiplicity] [: type] [= initial value]
[{property}]

- The *property* choices are **changeable**, **addOnly**, and **frozen**

Student Info
-id: string
-hasGraduated: bool = false

Full UML Operation Specification

- The full form of a UML operation is as follows:

[visibility] name [(parameter-list)] [:return-type] [{property}]

- The *property* choices are **sequential**, **concurrent**, **guarded**, and **isQuery**

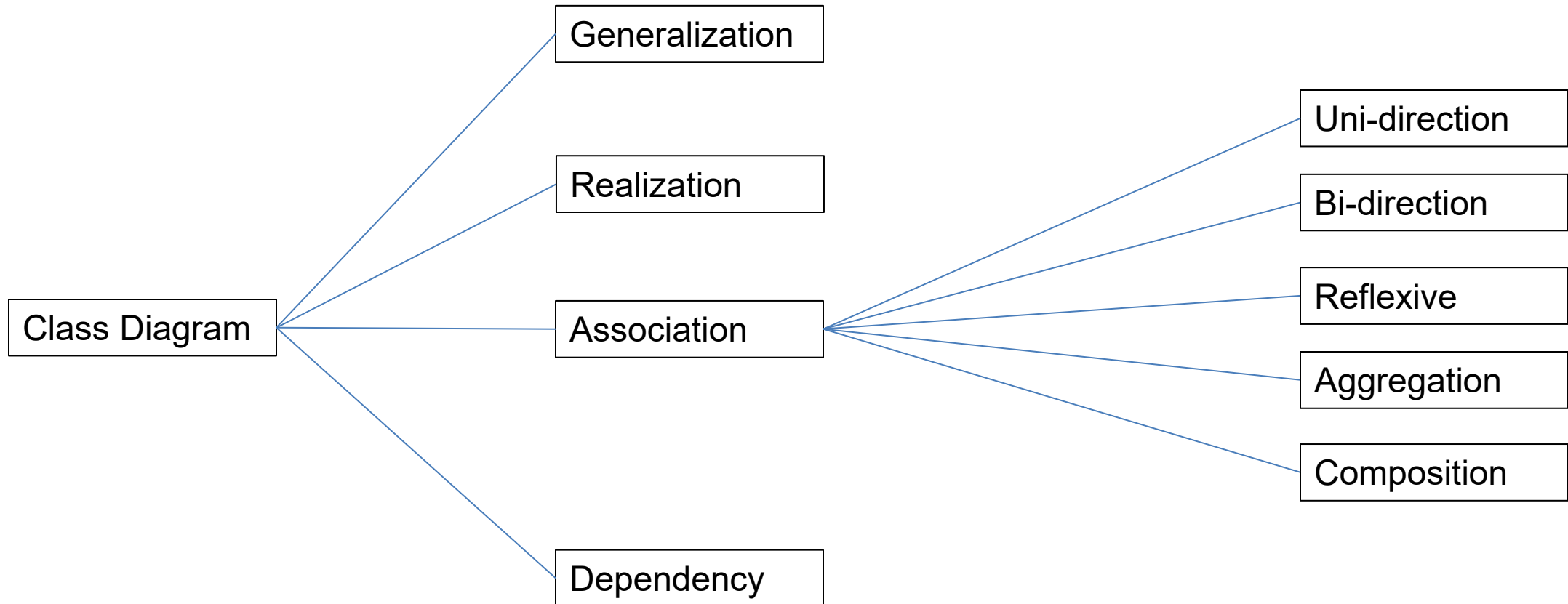
- The full form of a UML parameter is:

[direction] name : type [= default-value]

- The *direction* choices are **in**, **out**, and **inout**

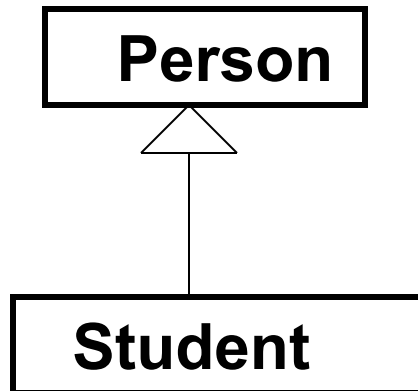
UML Class Relationships

- A class relationship is a connection between two (or more) classes
- The three most important class relationships are generalizations, associations, and aggregations
- UML provides a graphical representation for each of the relationships using a different line type for each relationship



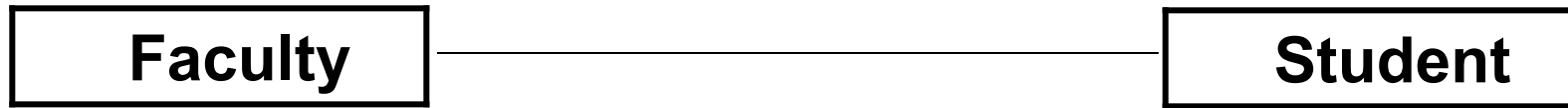
Generalization

- A generalization is a relationship between a general thing (superclass) and a more specific kind of that thing (subclass)
- In the UML, generalization requires that objects of the subclass may be used anywhere an object of the superclass appears



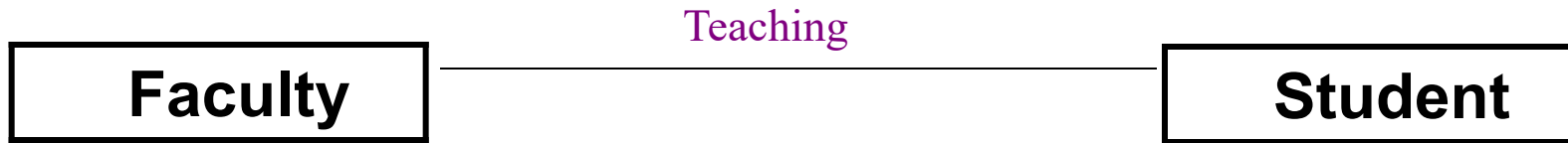
Association

- An association is a structural relationship that specifies that objects of one thing are connected to objects of another



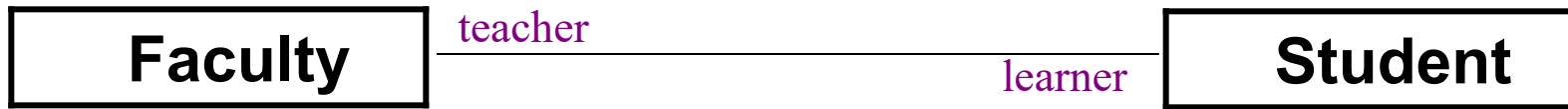
Association

- Associations can be adorned with a name.



Association

- Associations can be adorned with the roles.

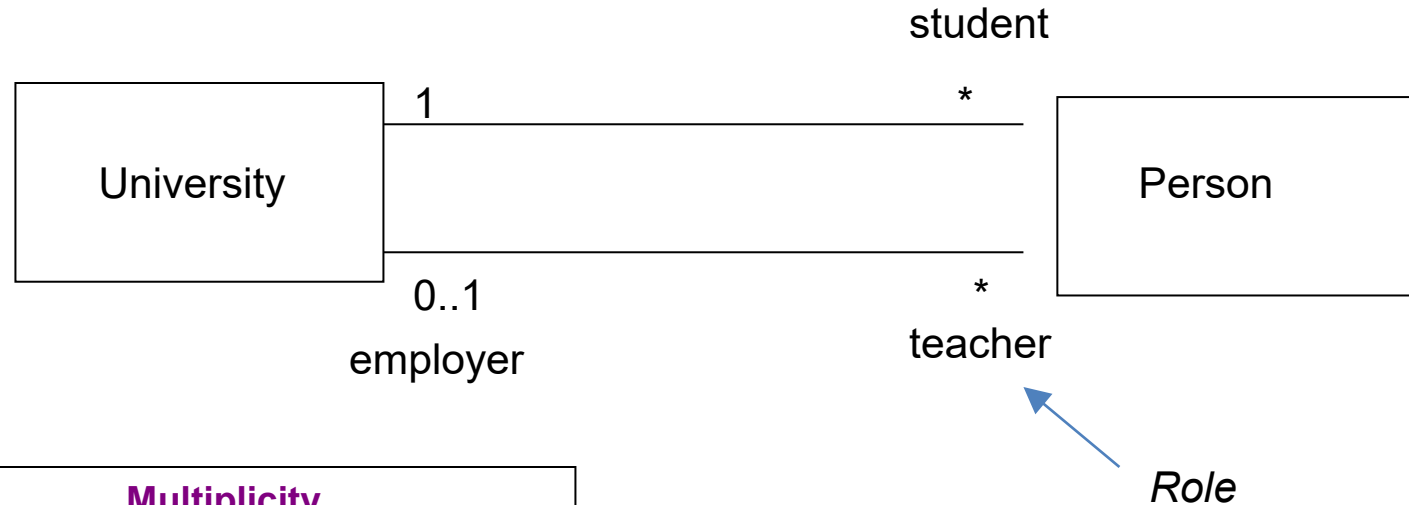


Association

- Associations can be adorned with the multiplicity.



Association: Multiplicity and Roles



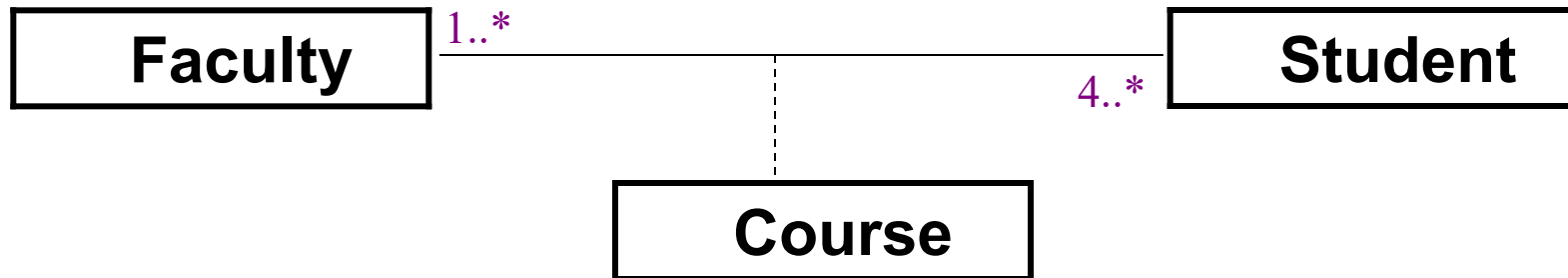
Multiplicity	
Symbol	Meaning
1	One and only one
0..1	Zero or one
M..N	From M to N (natural language)
*	From zero to any positive integer
0..*	From zero to any positive integer
1..*	From one to any positive integer

Role

“A given university groups many people; some act as students, others as teachers. A given student belongs to a single university; a given teacher may or may not be working for the university at a particular time.”

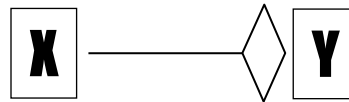
Association Class (relation attributes)

- Each object of association class is one instance of relationship (link) in an association.



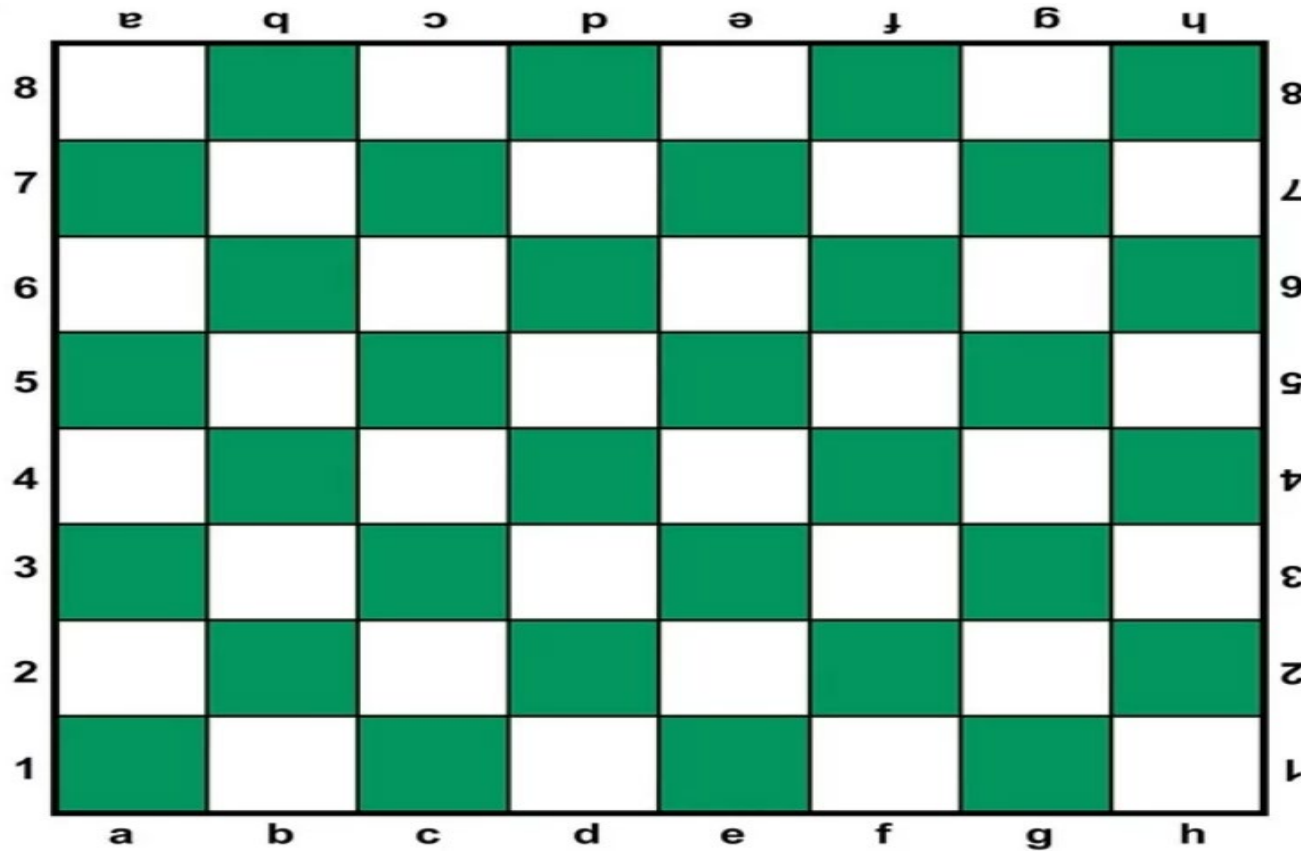
Aggregation

- An aggregation is an association that represents whole/part relationship
- The “whole” end of the association relationship is adorned with an open diamond shape (e.g. X is part of Y)
- e.g. door:Door is part of car:Car



Composition

- Models the part–whole relationship but, in addition, Every part may belong to only one whole, and If the whole is deleted, so are the parts



Generalization, Aggregation, and Association

- See if the following pairs of classes have generalization, aggregation or association
 - Faculty & student (as)
 - Hospital & doctor (as)
 - Door & Car (ag --<>)
 - Member & Organization (ag --<>)
 - People & student (ge <|--)
 - Circle & point (ge --|>)
 - Department & Faculty (as)
 - Employee & Faculty (ge <|--)
 - Item & Printer (ge <|--)
 - Account & Checking account (ge <|--)